OUADAHI MOHAND BELKACEM

Software & AI Engineer

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EDUCATION

Bachelor in Autonomous Systems and Ambient Software and Mobile AI *2023 – 2025*
Numidia Institute of Technology (NIT), Algiers, Algeria
Final Year GPA: 16.64 / 20 (4.0 / 4.0 equivalent)
Overall GPA: 14.36 / 20 (3.7 / 4.0 equivalent)

Computer Science (Transferred) *2019 – 2023*
Université M'hamed Bougara – Faculté des Sciences, Boumerdès, Algeria

PROFESSIONAL POSITIONS HELD

Research Intern – Synchromédia Laboratory *September 2026 – January 2027*
École de technologie supérieure (ÉTS), Montréal, Canada

- Project: Autonomous and Interpretable Decomposition of Degraded RGB Documents, supervised by Prof. Mohamed Cheriet and co-supervised by Saida Khazri.
- Challenge: a degraded manuscript mixes more physical layers (ink, paper, stains, bleed-through) than the 3 channels an RGB scan records, an underdetermined inverse problem with no unique solution.
- Extending the lab's PRISM model to RGB with sparsity, spatial and physics-informed constraints.
- Studying document physics, colour spaces and evaluation metrics toward a layer-level benchmark.

AI Engineer – Full Time *August 2025 – Present*
Numidia Institute of Technology (NIT), Algiers, Algeria

- Designing and building a multimodal AI medical assistant integrating Automatic Speech Recognition (ASR), X-ray/MRI imaging analysis (DenseNet-121, Xception, EfficientNet-B2), and OCR-based document automation.
- Implementing RAG-powered clinical decision support using ChromaDB vector database (1,200+ medical documents) with LLM orchestration (Qwen-72B, Mixtral-8x22B, DeepSeek-67B) for evidence-based diagnosis verification.
- Designing a privacy-first local inference architecture for secure medical document processing.

AI Engineer – Internship *May 2025 – August 2025*
Ooredoo Algeria, Algiers, Algeria

- Designed and deployed a personalized subscription recommendation system now live in production for Ooredoo Algeria's entire customer base.
- Built Big Data architecture using PySpark and Cloudera Hadoop to process 400+ subscriber data files of 40–50 GB each (16+ TB total), covering call destinations, traffic, and subscription data.
- Delivered a complete end-to-end ML pipeline: feature engineering, model benchmarking, hyperparameter tuning, and production deployment.

Full-Stack Developer & AI Engineer – Freelance *2025 – Present*
Private School Client, Algiers, Algeria

- Built a comprehensive school management platform covering enrollment, staff management, payroll processing, and parent communication portal.
- Integrated AI-powered facial recognition to automate staff attendance tracking and parent-detection for secure child pickup notifications.

Full-Stack Developer & AI Engineer – Freelance *2025*

Travel Agency Client, Algiers, Algeria

- Designed and developed a full-stack web application for travel program management, agency dashboard, and client portal with integrated third-party APIs.

COURSES TAUGHT AND SERVICES TO STUDENTS AND INSTITUTION

Technical Mentor & Jury Member *2023 – Present*

NCS Club – Numidia Institute of Technology

- Mentored student teams participating in national hackathons on topics including AI system design, full-stack development, and machine learning pipelines.
- Served as technical jury member evaluating student projects at the DevImpact Hackathon, providing structured feedback on technical quality and innovation.
- Organized and managed logistics for the DevImpact Hackathon, attracting participants from multiple Algerian higher-education institutions.

PUBLICATIONS

Manuscripts in preparation:

Multimodal Deep/Machine Learning for Pediatric Appendicitis Diagnosis *In preparation, 2026*

Ouadahi Mohand Belkacem [et al.] — Manuscript in preparation, target journal TBD.

Abstract: Pediatric appendicitis is a common surgical emergency where timely and accurate diagnosis is critical, yet ultrasound interpretation remains highly operator-dependent and single-modality models have shown limited clinical reliability. This work presents a complete multimodal deep and machine learning study for automated pediatric appendicitis diagnosis using the Regensburg Pediatric Appendicitis Dataset (782 patients, 2,097 ultrasound images, 58 clinical features). On structured clinical data, XGBoost achieves 94.23% accuracy and an ROC-AUC of 0.9746 with only 2 false negatives on the held-out test set. For ultrasound image classification, the proposed ScaleAware-Attention architecture achieves an AUROC of 0.8606, surpassing the published image-only state of the art by 0.06 AUROC points. A Deep Multimodal Fusion Network combining the vision module with a GANDALF-based tabular branch through cross-attention achieves an AUROC of 0.9609, a balanced accuracy of 87.27%, with 100% recall on healthy patients and 100% precision on appendicitis predictions — a +19.7 percentage point improvement in AUROC over the best image-only model.

No peer-reviewed publications yet. Research pipeline is active (see Projects & Research below).

OTHER PROFESSIONAL ACTIVITIES

Research Projects & AI Systems Development

- **AI-Powered Medical Assistant Pipeline (2025–Present):** End-to-end clinical decision support system combining ASR, OCR, and deep learning for medical imaging (DenseNet-121, Xception, EfficientNet-B2, custom CNN for Alzheimer's detection).
- **DiaryAI – AI-Powered Personal Journal App (2025–2026):** Multi-stage NLP pipeline with sentiment analysis, emotion classification, mood scoring, HealthKit biometric integration, and voice-to-insight pipeline. Tech: React Native, Gemini API, Supabase.
- **Handwritten Prescription OCR & Labeling System (2025–2026):** End-to-end OCR pipeline using CRAFT, TrOCR, and Tesseract with fuzzy medication database matching and Qwen LLM semantic classification. Designed for custom Algerian prescription model training.
- **BMI Prediction from Facial Images (2025):** CNN-based model exceeding accuracy benchmarks from prior published research on the same dataset — publication-quality performance via novel preprocessing, augmentation, and architecture strategies.
- **Advanced ASR System for Quran & Prayer Assistance (2025):** Custom speech recognition system with real-time mispronunciation detection; full MVP on Flutter / FastAPI / PostgreSQL stack.
- **PPG-Based Heart Rate Analysis & Dataset Generation (2026):** Smartphone-based photoplethysmography signal acquisition with real-time peak detection, synthetic waveform generation, and HRV-enriched AI-training-ready dataset pipeline.

Workshops & Hackathons

- Accepted into Junction Asia Hackathon (international) — one of the most competitive hackathon programs in Asia.
- Participated in multiple national Algerian hackathons, earning technical recognition.
- DevImpact Hackathon Organizer — designed challenge tracks and managed multi-institution participation.

MEMBERSHIP AND ACTIVITIES IN PROFESSIONAL ASSOCIATIONS

External Relations Manager – NCS Club *2023 – Present*
Numidia Institute of Technology, Algiers, Algeria

- Built institutional partnerships and managed external collaborations with organizations, companies, and universities.
- Represented NIT at national and international events and competitions.
- Coordinated with external stakeholders for sponsorships and technical event logistics.

PROFESSIONAL HONORS, AWARDS, AND FELLOWSHIPS

- **Academic Excellence – Final Year GPA:** 16.64 / 20 (4.0 / 4.0 equivalent), Numidia Institute of Technology, 2025.
- **IELTS Band 7 (Fluent English):** Achieved international certification of English-language proficiency.
- **Cisco Networking Certification:** Professional certification in networking fundamentals and infrastructure.
- **Junction Asia Hackathon Acceptance:** Selected for one of the most competitive international hackathon programs — recognizes outstanding technical and innovative capacity.
- **Research Achievement:** BMI prediction model from facial images surpassing benchmarks reported by prior published research on the same dataset — results of publication-ready quality.
- **Production Deployment at Ooredoo Algeria:** ML recommendation system deployed at national scale for Algeria's leading telecommunications provider.

COMMUNITY SERVICE

- **DevImpact Hackathon Organization:** Designed and organized a multi-institution hackathon with the goal of fostering AI and software engineering talent among Algerian university students.
- **Technical Mentorship:** Provided ongoing mentorship to junior students at NIT on AI project design, career guidance, and technical skill development.
- **Community Engagement:** Active participant in impactful national technology initiatives and community-oriented events aimed at advancing Algeria's tech ecosystem.

Technical Skills (Summary)

Languages: Python, JavaScript (React.js, Next.js, Node.js), Swift, C, HTML/CSS **Mobile:** Flutter, React Native

ML / AI: TensorFlow, PyTorch, Scikit-learn, OpenCV, HuggingFace Transformers, Riva ASR, MLlib **Big Data:** PySpark, Cloudera Hadoop

Research Areas: Computer Vision, Document Image Analysis, Medical Imaging, Matrix Factorization (NMF), Inverse Problems

Databases: PostgreSQL, MongoDB, ChromaDB, Supabase **APIs:** Google Gemini, Groq, HuggingFace Router, FastAPI
Design: Figma

Languages Spoken: Arabic (Native), English (Fluent – IELTS Band 7), French (Fluent)

Reference

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